

Entry and Advertising on Digital Platforms

Evidence from a Large E-Commerce Platform

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Motivation: Research Agenda

- Information Asymmetry + Empirical IO

- ▷ How car insurers compete when they have different information about consumers' riskiness

1. Jin and Vasserman ('25): buying data from consumers via monitoring programs (JPE)
2. Jin and Yu ('22) & Jin ('25): moral hazard and “first-best” contracts with telematics data (EC)
3. Cosconati et al. ('25): competition with information heterogeneity (EC + ReStud)

- ▷ How e-commerce platforms manage seller entry with little quality information vs. incumbents

1. Jin and Sun ('19): platform as data vendor to sellers
2. Jin and Sun ('22): AI training for new sellers
3. This paper: entry and advertising

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Millions of new businesses join digital platforms each year

- In 2020, around 1.3 mil new Amazon sellers; 6.8 mil for Taobao; 7 mil new channels (5+ subscribers) for Youtube (Marketplace Pulse; Socialblade)

- Success rates are largely unknown despite anecdotal accounts of rising concentration

“The top 1% of sellers have been selling on Amazon for years and drive at least half of the sales volume. The thousands that join daily don't have the same experience, many never get a single order.” (Marketplace Pulse, 2024)

“New entrants [on Tiktok Shop] remain restricted from reaching creators... It's a classic catch-22: you need affiliates (creators) to grow, but you can't access them until you've already grown.” (Perez-Tenessa for Trendio, 2025)”

⇒ a “cold-start” problem (Tadelis, 2016; Milgrom and Tadelis, 2018)

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Motivation: Gap in Literature

Two main mechanisms to assess entrant quality:

- *explore* by steering consumers to sample
- solicit bids via *advertising* (Chen and He 2011; Athey and Ellison 2011)

These mechanisms have become constrained as entry volume grows

- exogenous entrant sampling → higher growth and survival (Pallais 2014; Bai et al. 2020)
- many entrants rationally bid below (or not pay) the cost for reputation (Gao and Yu 2023; Aleksenko and Kohlhepp 2023; Cui 2025; Dendorfer and Seibel 2025)
- established businesses advertise far more than entrants, sometimes with higher returns (Dai et al. 2023; Moshary 2021; Sahni and Zhang 2023)
- findings draw on businesses that already earned revenue—small fraction of potential entrants
- A systematic account of entry outcomes on major platforms is still absent

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Research Questions & Findings

This paper

- Rich admin. data from a major Chinese e-commerce platform
- ? How likely is it for an entrant to survive on the platform?
- ? How does the platform explore entrant quality given the “cold-start problem”?
- ? Can more advertising promote entry?
- Propose a theory of platform-mediated entry to jointly understand how baseline screening via organic ranking, (bandit) exploration, and advertising shape entry

Research Questions & Findings

This paper

- Rich admin. data from a major Chinese e-commerce platform
- ? How likely is it for an entrant to survive on the platform?
define and describe a narrow “entry funnel”
- ? How does the platform explore entrant quality given the “cold-start problem”?
construct consideration-set data, entrants are subject to more intense screening than incumbents and are under-explored relative to social optimum.
- ? Can more advertising promote entry?
revisit an RCT that raised advertising (+2.3%): promoted entry with minimal impact on screening, but only 7% of new sellers ever advertised.
- Propose a theory of platform-mediated entry to jointly understand how baseline screening via organic ranking, (bandit) exploration, and advertising shape entry

Related Literature

◦ Entry on digital platforms

- ▶ Earlier work focuses on network effects / discoverability without quality uncertainty (Rochet and Tirole 2003; Armstrong 2006; Hagiu and Wright 2024)
- ▶ Exploration, reputation, and optimal subsidy optimal subsidies (Bai et al. 2025; Cui 2025; Dendorfer and Seibel 2025; Aleksenko and Kohlhepp 2023; Gao and Yu 2023)
- ▶ Novel use of consideration-set data (Honka et al. 2024; Ursu 2025; Lam 2025; Yu 2025)

◦ Optimal exploration

- ▶ Platforms should “start small,” scaling exposure as credibility builds (Che and Hörner 2018)
- ▶ Information design around reviews (Vellodi 2018; Papanastasiou et al. 2018)

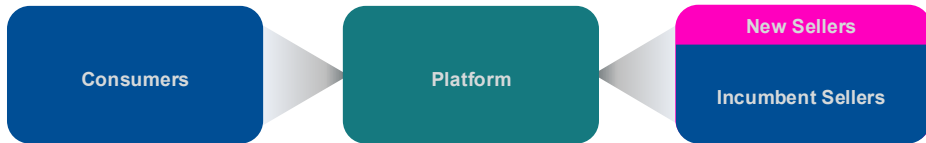
◦ Advertising and entry

- ▶ Advertising can substitute for past consumer experience (Shapiro 1982)
- ▶ Mixed evidence on signaling value that “may or may not apply to small and new entities” (Blake et al. 2015; Sahni and Nair 2020; Moshary 2021; Sahni and Zhang 2023)
- ▶ Interaction with other platform design choices (Bar-Isaac and Shelegia 2022; Yu 2025)

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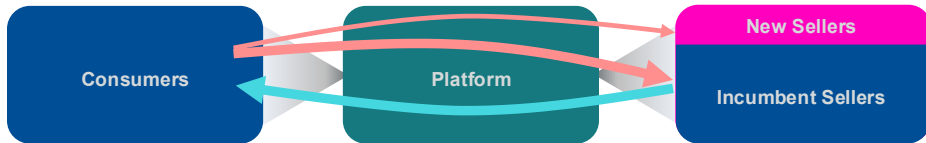
A Large E-Commerce Platform

- millions of active and independent sellers (mostly retailers), hundreds of millions of consumers: a large share of Chinese e-commerce retail in 2019 (emarketer 2018)



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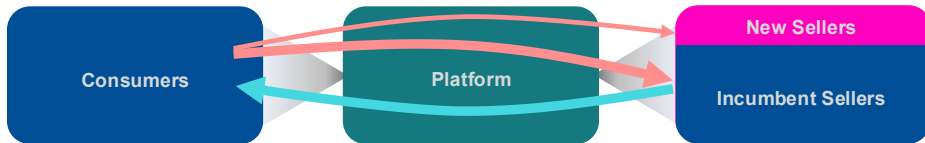


platform matches consumers and sellers through

- ▶ **search** query → keywords → rank related products (algo + auction)
- ▶ **recommendation** past consumer choice and seller conversion (algo)

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consumers engage in **costly searches**

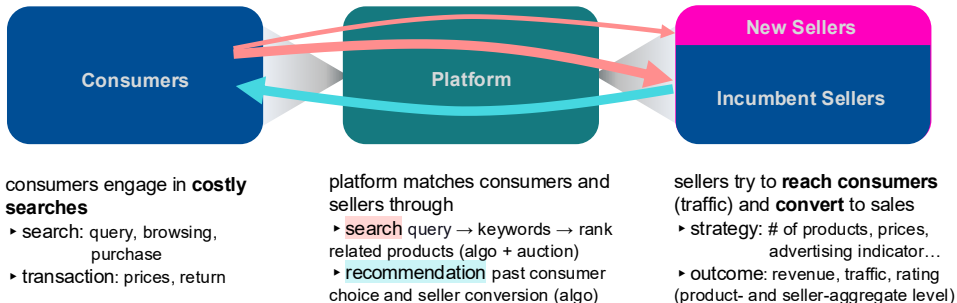
- search: query, browsing, purchase
- transaction: prices, return

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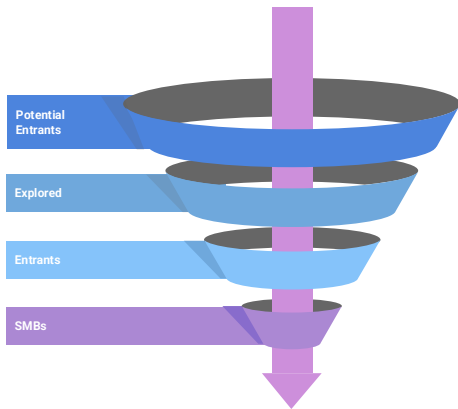
Two Datasets

- Aggregate entry statistics spanning 2015 to 2019
 - ▷ Visualize the entry funnel and its evolution as the platform matured
- Micro-data from a prior RCT covering all entrants from May - August, 2019
 - ▷ consideration-set panel by merging:
 - * search-session records (consumer/product IDs)
 - * seller and product characteristics
 - * transaction data
 - ▷ seller-level randomization; treatment → more advertising
 - ▷ data limitation due to the platform policy: no search rankings or ad bids.

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Entry Funnel

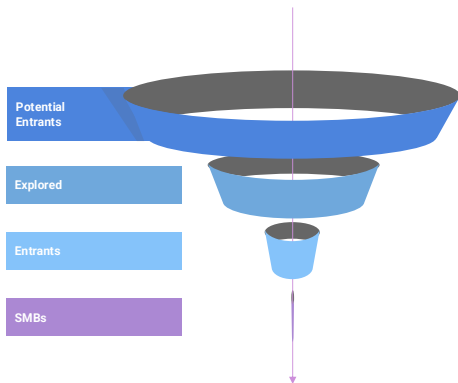
Funneling a cohort of new sellers.



- **Potential Entrants / Registrants:** all seller registrations (25K/day in 2019)
- **Explored:** Registrants with average monthly traffic above 0: ~51%
- **Entrants / New Sellers:** Registrants with listings and positive revenues: ~15%
- **New SMBs:** Entrants earning meaningful revenues (90th pctl of all sellers): 0.6% in month 1; 0.4% in month 12
- Most prior studies focus on SMBs or above; our results below are primarily driven by entrants (Dubé 2021)

Entry Funnel

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Entry Dynamics

Table: Entry Funnel from 2015 to 2019

Year	2015	2016	2017	2018	2019
<i>Entry Funnel</i>					
Explored (0+ UV) per 10K P.E.	3246	3917	4536	4552	5069
Explored (100+ UV) per 10K P.E.	2791	3164	3001	2868	2280
Entrants per 10K P.E.	1940	2296	2953	2693	1525
SMB (month 1) per 10K P.E.	87	107	125	124	62
SMB (month 12) per 10K P.E.	59	55	55	51	43
<i>Entry Volume (vs. 2019)</i>					
Total Number of P.E.	1.33x	1.17x	0.86x	0.75x	1x
Number of Explored P.E.	0.85x	0.91x	0.77x	0.67x	1x
Number of Entrants	1.69x	1.76x	1.66x	1.32x	1x
Number of SMBs (month 1)	1.88x	2.04x	1.74x	1.51x	1x
Number of SMBs (month 12)	1.83x	1.49x	1.09x	0.90x	1x

Notes: P.E. standards for “potential entrants” and UV for the total number of “unique visitors” ever received. This table is derived based on aggregated administrative data from the platform, with entry volume numbers benchmarked against the 2019 values of the respective rows: e.g. in 2015, the total number of P.E. is 1.33x, which means that it is 133% the 2019 total number of P.E.

Entry Funnel and Dynamics

- The entry funnel has always been narrow and has become narrower over time.
- More potential entrants were explored, but the exploration has become shallower
- As fewer entrants became SMBs, those that did were more likely to survive (as SMBs)

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Platform Exploration

- Platform exploration is ultimately consumer search steered by the platform
 - we measure the degree to which entrants are sampled and the match quality of those sampled, in equilibrium
- Key empirical challenge: demand heterogeneity—consumers who search entrants may differ in unobservable ways from those who search incumbents
- We use *consideration-set data* for apples-to-apples comparisons:
 - ▷ purchase likelihood *within* consideration sets containing both new and incumbent sellers
 - ▷ purchase likelihood *across* nearly identical sets—matched on consumer, keywords, session size, and date—that differ in whether they include new sellers
- We do not observe search rankings or ad prices → goal is to identify the *state* of entrant under-exploration, not structurally separate platform steering from consumer search

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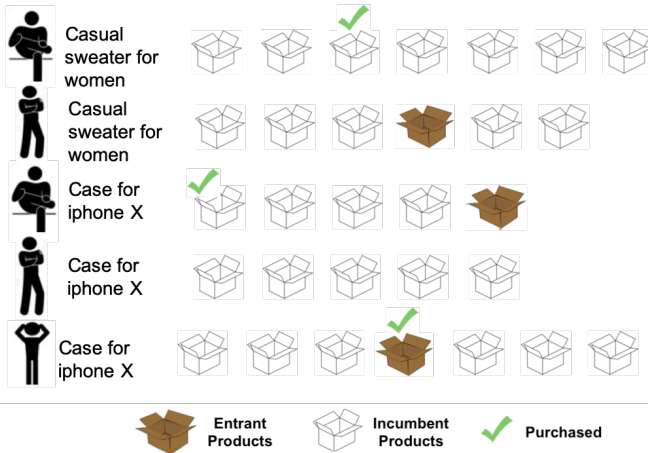
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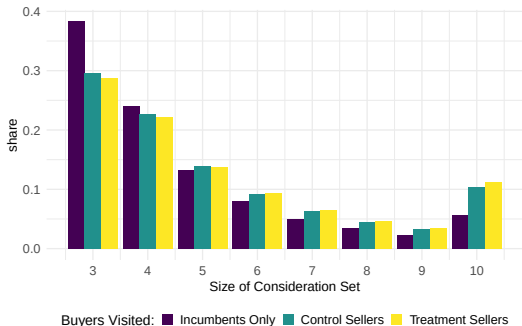
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Consumer Search Sessions - Consideration Set Data



- use clickstream data during search sessions to construct consideration sets
- each includes one consumer, a set of keywords extracted from search queries, and a set of products clicked (and “considered”) by the consumers and the corresponding sellers.
- 18% of consideration sets resulted in a purchase

Consumer Search Sessions - Consideration Set Data



- subsampling¹
 1. get consumers that visited entrants
 2. subsample among all of their search sessions
 3. identify popular search keywords
 4. for consumers that never visited entrants, subsample their search sessions with popular keywords
- focus on search sessions with 3+ sellers due to data contamination from “passive searches” (Ursu et al. 2022) and test/miscellaneous searches with vanishingly low conversion rates.
- search sessions with new sellers are larger – raw conversion rates comparisons mask demand heterogeneity, not “apples-to-apples” (Ursu 2018)

¹We have 10+ mil search sessions per day. [detail](#)

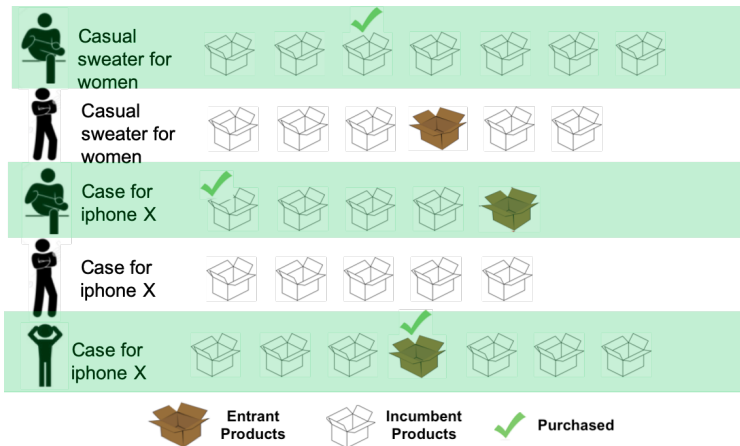
Consumer Search Sessions - Consideration Set Data

Table: Summary Statistics: Consideration Set Sample

Statistic	Mean	St. Dev.	Min	Pctl(25)	Pctl(99)	Max
Number of Sellers Browsed	4.6	2.7	3	3	12	55
% of New Sellers	0.57	8.7	0	0	33	100
Purchased (on the same day)	0.180	0.167	0	0	0	1
Purchased from New Sellers	0.001	0.033	0	0	0	1
Pay Amount (USD)	4.8	21.2	0	0	753	2,050
Avg. Seller Price Level (USD)	34.4	86.3	0.0	8.8	582	9,674
Avg. Seller Rating	13.0	2.9	0	11.5	20	20

Notes: $N = 6,007,329$. Unit of analysis is a consideration set. Dollar amount converted to USD based on the average 2019 currency exchange rate of 1 RMB to 0.1448 USD.

1. Match Quality Conditional on Search (Y = conversion rate)



- focus on search sessions with purchases → realized “market” (highlighted in green)
- comparing entrant vs. incumbent conversion rates within consideration sets
- consideration-set FEs to control for demand heterogeneity²

²We estimated seller-level brand effects with BLP and several price instruments [detail](#)

1. Match Quality Conditional on Search (Y = conversion rate)

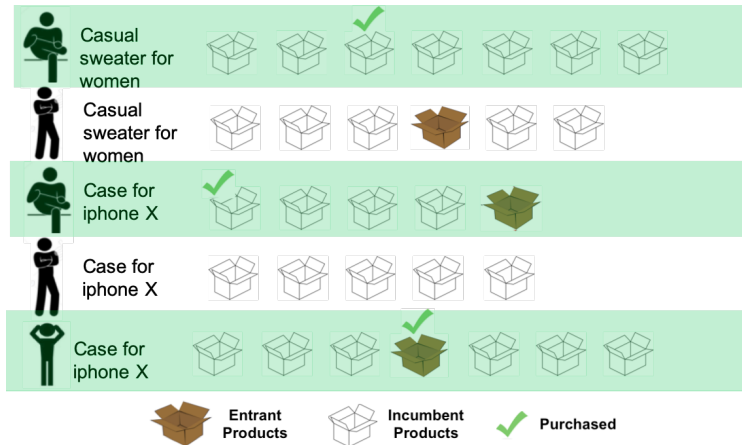
Table: Regression Results: Purchase Conditional on Search

	Has Purchased		Log Spending	
	(1)	(2)	(3)	(4)
Is New Seller	0.013*** (0.001)	0.003*** (0.001)	0.056*** (0.004)	0.010*** (0.004)
Incumbent Mean	0.17		0.71	
Seller Controls	N	Y	N	Y
Consideration-Set FEs	Y	Y	Y	Y
R ²	0.053	0.073	0.054	0.078
Adj. R ²	0.036	0.057	0.037	0.062

Note: $N = 1,084,228$ consideration sets after filtering down to those that contained at least one new seller. Reports estimates from Equation ???. All variables are described in the paragraphs above, with columns corresponds to dependent variables (labeled at the top) and model specifications (labeled at the bottom). * $p < .1$, ** $p < .05$, *** $p < .01$.

- entrants had 1.2pp or 7% *higher* purchasing probability than incumbents

2. Match Quality Conditional on Search (Y = refunds/repeat purchase)



- focus on search sessions with purchases
- compare return and repeat purchase rates across consideration sets (purchase from entrants vs. from incumbents)
- use consumer FE + keyword-size-date interacted FE to control for demand heterogeneity

2. Match Quality Conditional on Search (Y = refunds/repeat purchase)

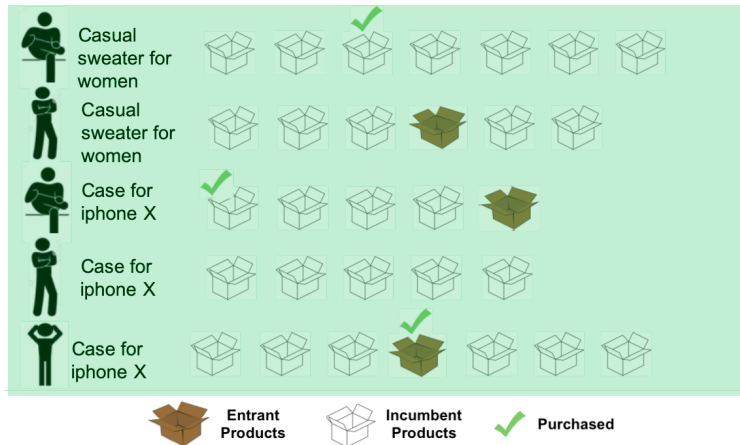
Table: Regression Results: Post-Purchase Experience

	<i>Dependent variable:</i>			
	Is Returned		Has Repeat Purchased	
	(1)	(2)	(3)	(4)
Has New Seller	-0.006*** (0.002)	-0.004*** (0.001)	0.006*** (0.002)	0.004** (0.002)
Incumbent Mean	0.078		0.064	
Seller Controls	N	Y	N	Y
Consumer FEs	Y	Y	Y	Y
Keyword-Size-Date FEs	Y	Y	Y	Y
R ²	0.398	0.463	0.301	0.378
Adj. R ²	0.181	0.257	0.050	0.082

Note: $N = 289,259$ consideration sets. We first filter down to those that resulted in a purchase; we then filter down to purchases in which the seller appeared at least one more time in the consumer's future consideration sets. The table reports estimates from Equation ?? . All variables are described in the paragraphs above, with columns corresponds to dependent variables (labeled at the top) and model specifications (labeled at the bottom). * $p < .1$, ** $p < .05$, *** $p < .01$.

- again, entrants had better match quality than incumbents
- next step: is this entry barrier socially inefficient?

3. Market Expansion Effects



- focus on all search sessions: from potential market to realized market
- compare across consideration sets with and without new sellers
- use consumer FE + keyword-size-date interacted FE to control for demand heterogeneity

3. Market Expansion Effects

Table: Regression Results: Market Expansion Effects

	<i>Dependent variable:</i>			
	Has Purchased		Log Spending	
	(1)	(2)	(3)	(4)
Number of New Sellers	0.003*** (0.001)	0.002*** (0.001)	0.011*** (0.003)	0.008*** (0.003)
Incumbent Mean	0.041		0.17	
Seller Controls	N	Y	N	Y
Consumer FEs	Y	Y	Y	Y
Keyword-Size-Date FEs	Y	Y	Y	Y
R ²	0.053	0.073	0.054	0.078
Adj. R ²	0.036	0.057	0.038	0.062

Note: $N = 6,007,329$ consideration sets. The table reports estimates from Equation ???. All variables are described in the paragraphs above, with columns corresponds to dependent variables (labeled at the top) and model specifications (labeled at the bottom).

* $p < .1$, ** $p < .05$, *** $p < .01$.

- across almost identical search sessions, replacing incumbents with entrants on the margin leads to market expansion.

Entrant Under-Exploration

- Despite receiving very little traffic, entrants are subject to more intense screening than incumbents: they convert at higher rates within the same consideration set
 - Inconsistent with bandit exploration, which would yield *lower* entrant conversion
- Marginal reallocation of search from incumbents to entrants expands the market
 - Entrants are under-explored compared to social optimum
- What about signaling?
 - ▷ advertising (sponsored listings) is by far the most prevalent
 - ▷ other examples: reward-for-feedback (Li, Tadelis, and Zhou 2019), limited-time promotional events/festivals

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Did in-platform advertising promote entry?

- In May - Aug 2019, Jin and Sun (2022) ran a platform-wide experiment giving a fraction of all entrants a digital training module.
- Advertising increased (+2.3% ITT on extensive margin) with precise null effects on other main seller strategies such as pricing and product variety.
- Generated +1.3% ITT on traffic.
- Caveats
 - ▷ Only 7% of new sellers ever advertised
 - ▷ Short-video product description also increased, but we found no contemporary increase in organic traffic.

ITT Effects on Entry Funnel Stages

Table: Intent-to-Treat Effects on Entry Funnel Stages

	<i>Dependent variable:</i>				
	Has Traffic (1)	Has Traffic (100+ UV) (2)	Has Revenue (3)	SMB (Mo. 1) (4)	SMB (Mo. 9) (5)
Treatment	-0.0003 (0.0020)	0.0028*** (0.0010)	0.0011** (0.0004)	0.00013 (0.00009)	0.00012** (0.00005)
Dep Var Mean	0.498	0.230	0.161	0.005	0.004
Adj. R ²	0.172	0.176	0.181	0.264	0.268

Notes: $Y_{ics} = \beta \text{Treatment}_i + \alpha_c + \alpha_s + \epsilon_{ics}$; cohort and sector FEs; SEs clustered by seller. * 10%, ** 5%, *** 1%.

- Per 10,000 P.E.: +28 with meaningful traffic, +11 with positive revenue +1–2 SMBs
- Bottleneck is low (7%) participation

1. Match Quality Conditional on Search (Y = conversion rate)

Table: Regression Results: Purchase Conditional on Search (Treatment - Control T test)

	Has Purchased		Log Spending	
	(1)	(2)	(3)	(4)
Is New Seller (Treatment - Control T test)	-0.0016 (0.0013)	-0.00081 (0.0013)	-0.0074 (0.0060)	-0.0039 (0.0059)
Seller Controls	N	Y	N	Y
Consideration-Set FEs	Y	Y	Y	Y

Note: $N = 1,084,228$. See notes from Table ?? for more details. * $p < .1$, ** $p < .05$, *** $p < .01$.

- no detectable differences; marginal entrants reached through advertising were slightly lower quality but remained substantially superior to incumbents

2. Match Quality Conditional on Search (Y = refunds/repeat purchase)

Table: Regression Results: Post-Purchase Experience (Treatment - Control T test)

	<i>Dependent variable:</i>			
	Is Returned		Has Repeat Purchased	
	(1)	(2)	(3)	(4)
Has New Seller (Treatment - Control T test)	0.0054** (0.0026)	0.0026* (0.0016)	0.0024 (0.0024)	0.0019 (0.0015)
Seller Controls	N	Y	N	Y
Consumer FEs	Y	Y	Y	Y
Keyword-Size-Date FEs	Y	Y	Y	Y

Note: $N = 289,259$. See notes from Table ?? for more details. * $p < .1$, ** $p < .05$, *** $p < .01$.

- slightly increased refund rates ($p < .05$); marginal entrants were slightly lower quality
- directionally reduced the exploration gap for repeat purchase; screening efficiency largely preserved

3. Market Expansion Effects

Table: Regression Results: Market Expansion Effects

	<i>Dependent variable:</i>			
	Has Purchased		Log Spending	
	(1)	(2)	(3)	(4)
Number of New Sellers	-0.00046 (0.00044)	-0.00023 (0.00044)	-0.0025 (0.0020)	-0.0018 (0.0020)
Seller Controls	N	Y	N	Y
Consumer FEs	Y	Y	Y	Y
Keyword-Size-Date FEs	Y	Y	Y	Y

Note: $N = 6,007,329$. See Table ?? for more details. * $p < .1$, ** $p < .05$, *** $p < .01$.

- no detectable differences; advertising promoted entry with minimal impact on screening efficiency

- 1 Setting and Data
- 2 Entry Funnel and Dynamics
- 3 Are Entrants Under-Explored?
- 4 Can More Advertising Promote Entry?
- 5 A Theory of Platform-Mediated Entry

Baseline: Platform-Mediated Search

Consider a search session with $J \geq 1$ incumbent products, each with price-inclusive quality $\delta_j \in [0, 1]$ (Athey & Ellison 2011)

- Platform knows δ_j and ranks (descending) all J incumbents
- Consumers search sequentially with per-inspection cost $s \sim U[0, 1]$; match with probability δ_j ; stop when expected benefit falls below s
- Equilibrium quality and traffic at rank r :

$$\theta^B(r, J) = \frac{J - r + 1}{J + r}, \quad \varphi^B(r, J) = \bar{s}^B \cdot \prod_{r' < r} (1 - \theta^B(r', J))$$

All consumers with $s < \bar{s}^B = \theta^B(1, J) = J/(J+1)$ will search

- Consumer welfare proof:

$$W^B(J) = \frac{1}{2\bar{s}^B} \sum_{r=1}^J \varphi_r^B (\theta_r^B)^2$$

Timing

1. Nature draws $\delta_j \sim U[0, 1]$
2. Platform ranks by quality
3. Consumers search

Information

- ▶ Platform knows δ_j
- ▶ Consumers learn δ_j only upon inspection

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Timing

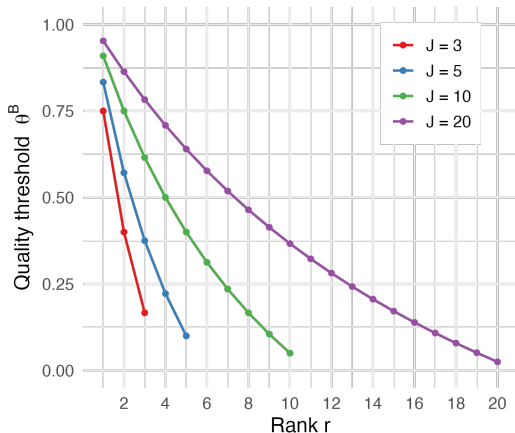
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Baseline Equilibrium: Quality Cascade

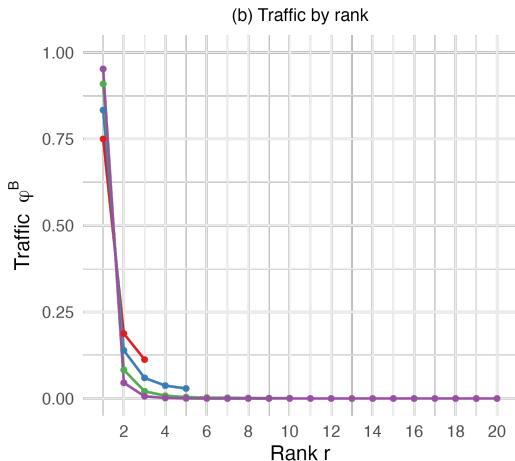
(a) Quality thresholds



$$\theta^B(r, J) = \frac{J - r + 1}{J + r}$$

- ▷ Expected quality at rank r among J incumbents (Athey & Ellison 2011)
- ▷ Entry threshold: $\bar{s}^B = \theta_1^B = \frac{J}{J+1}$
- ▷ Quality declines with rank; steeper for larger J
- ▷ Consumers stop searching when cost exceeds expected benefit $\rightarrow$ consideration set.

Baseline Equilibrium: Traffic Cascade



$$\varphi^B(r, J) = \bar{s}^B \cdot \prod_{r' < r} (1 - \theta^B(r', J))$$

- ▶ Traffic at rank r : fraction of consumers who reach r and inspect
- ▶ Concentrated at top positions; decays sharply with rank
- ▶ Larger $J \rightarrow$ steeper hierarchy $\rightarrow$ more at stake from displacement

Entrant and Platform Exploration

A single entrant with unknown quality δ . Platform knows the distribution but not δ ; it learns only as consumers inspect δ .

- $\delta \sim \text{Beta}(2q, 2-2q)$, with mean $q \in (0, 1)$
 - ▷ $q = 0.5 \Rightarrow U[0, 1]$
- Platform *explores* by top-ranking the entrant at rate $\kappa \in [0, 1]$.
- Three configurations:
 - ▷ **Exploration** (prob κ): entrant at rank 1, displaces incumbents one rank deeper
 - ▷ **Organic** (prob $1-\kappa$): entrant at rank r_{org} based on prior q
 - ▷ **Full information** (benchmark): entrant at true rank $r^*(\delta)$
- $W^E(\delta, J)$, $W^O(\delta, J)$ **linear** in δ ; $W^*(\delta, J)$ **convex** proof
 - ▷ Rank fixed $\Rightarrow \delta$ and $(1-\delta)$ enter linearly; under full info, rank depends on $\delta \Rightarrow$ convexity

Timing

1. Nature draws δ
2. Platform sets κ , ranks
3. Consumers search
4. Platform updates belief

Information

- ▷ Platform and consumers know δ only upon inspection by consumers

Cold-start

- ▷ $q < \theta_1^B \Rightarrow \varphi_{\text{org}}^O \rightarrow 0$

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Marginal Learning Cost and Benefit

Since $W^E(\delta, J)$ and $W^O(\delta, J)$ are linear in δ :

$$\text{MLC}(J, q) = W^O(q, J) - W^E(q, J) \geq 0$$

$$L(J, q) = \mathbb{E}_\delta [W^*(\delta, J)] - W^O(q, J) \geq 0$$

Proposition 1 (Diminishing Returns to Exploration)

The per-session welfare gain from exploration is

$$W(\kappa) - W^O = \kappa(\text{MLB}(\kappa) - \text{MLC})$$

where $\text{MLB}(\kappa) \equiv \beta \cdot L \cdot \varphi_{\text{org}}^O / T(\kappa)$ is the marginal learning benefit and $T(\kappa) \equiv \varphi_{\text{org}}^O + \kappa(\bar{s}^E - \varphi_{\text{org}}^O)$ is total entrant traffic. A unique $\kappa^* \in (0, 1]$ exists whenever $\text{MLB}(0) = \beta \cdot L > \text{MLC}$.

proof

- ▷ **MLC**: welfare loss from displacing top incumbent with unknown entrant; $\uparrow$ in J , $\downarrow$ in q
- ▷ **MLB**(κ) = $\beta \cdot L \cdot \varphi_{\text{org}}^O / T(\kappa)$:
 - ▷ β : learning speed $\times$ future demand
 - ▷ L : learning gap (Jensen bonus from W^* convexity)
 - ▷ $\varphi_{\text{org}}^O / T(\kappa)$: diminishing returns—posterior precision grows linearly with entrant traffic; organic baseline φ_{org}^O already provides information, so marginal value of exploration decays as κ rises

Aggregate Welfare and Optimal Exploration

Platform chooses uniform κ across search sessions

$$\arg \max_{\kappa} \mathcal{W}(\kappa) \equiv \sum_{J,q} w(J,q) \cdot W(\kappa; J,q).$$

Proposition 2 (Exploration Dilutes Baseline Screening)

The aggregate entrant conversion rate $\bar{C}_E(\kappa)$ is decreasing in κ .

proof At $\kappa = 0$, organic ranking screens entrants by concentrating traffic on high- q sessions. As κ rises, traffic shifts toward low- q sessions, diluting the aggregate conversion rate.

Proposition 3 (Optimal Exploration)

$$\frac{\partial \kappa^*}{\partial \beta} \geq 0, \quad \frac{\partial \kappa^*}{\partial \alpha} \geq 0, \quad \frac{\partial \kappa^*}{\partial \mu} \geq 0$$

proof As platform matures (α falls, J increases): MLC rises, κ^* declines $\rightarrow$ narrowing entry funnel.

Three primitives govern κ^* :

Within-session convexity

β learning amplifier
per-inspection learning
 $\times$ expected future demand
higher $\beta \rightarrow$ learning $\uparrow$

Across-session composition

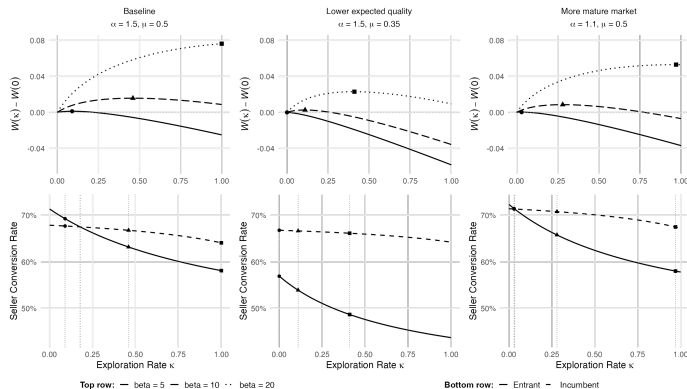
α category maturity

$J \sim \text{Pareto}(\alpha)$ **A8**
lower $\alpha \rightarrow$ more mature

μ avg. entrant quality

$q \sim \text{Beta}(2\mu, 2-2\mu)$
higher $\mu \rightarrow$ higher q (expected δ) $\rightarrow$ MSC $\downarrow \rightarrow \kappa^* \uparrow$

Simulation: Welfare and Conversion Rates



- **Top:** welfare gain $W(\kappa) - W(0)$ for $\beta \in \{5, 10, 20\}$

Prop. 3: κ^* reduces (dots shift left) as $\mu \downarrow$, platform matures, or β reduces

- **Bottom:** entrant vs. incumbent conversion rates

Prop. 2: entrant conversion $\downarrow$ in κ

At low κ : entrant can outperform incumbent due to organic screening alone

Frictionless Advertising

Platform offers rank-1 placement via posted price a^* , inducing a screening threshold δ^* : only entrants with $\delta \geq \delta^*$ are promoted.

Proposition 4 (Unraveling of Exploration)

Frictionless advertising dominates exploration: $\kappa^* = 0$, with screening threshold $\delta^*(J, q) > q$.

proof

Entrant's WTP for rank 1 (under liquidity constraint):

$$v(\delta) = (1 - \kappa)(\bar{s}^E - \varphi_{\text{org}}^O) \cdot \delta$$

Monotone in $\delta \Rightarrow$ posted price a^* induces cutoff δ^* .

Assumptions

- ▶ Liquidity constraint: entrant bids at most current-period payoff (Gao and Yu 2023; Cui 2025)
- ▶ No-disclosure: consumers observe ranking but not promotion mechanism
- ▶ Including pricing choice ($p^* = \delta'/2$) scales $v(\delta)$ by δ ; monotonicity preserved (Aleksenko and Kohlhepp 2023; Dendorfer and Seibel 2025)

Information Frictions

Fraction $\lambda \in [0, 1]$ of entrants participates in advertising; non-participants fall back to κ mechanism.

- Among participants, only $\delta \geq \delta^*$ promoted
- For non-participants, exploration trade-off identical to Section 4.2

Proposition 5.1 (Information Frictions Restore Exploration)

Suppose $\kappa_{\text{no-ads}}^* > 0$. Then $\kappa^* > 0$ whenever $\lambda < \bar{\lambda}$, where:

$$\bar{\lambda} = \frac{\mathbb{E}_{\delta}[\Delta W(\delta)]}{\mathbb{E}_{\delta}[\Delta W(\delta) \cdot 1(\delta \geq \delta^*)]} \in (0, 1)$$

▷ On our platform, only **7%** of new sellers had ever submitted a budget for sponsored listings → low λ

- ▷ Numerator: net gain from promoting *all* types
- ▷ Denominator: net gain from above-threshold only

Incumbent Blocking Restores Exploration

Incumbent blocking: second-price position auction.

Rank-1 incumbent's value of retaining position:

$$v_{\text{inc}}(q, J) = \underbrace{\theta_1^B [\bar{s}^B - \bar{s}^E (1-q)]}_{\text{static displacement}} + \underbrace{\beta_I \int_{\theta_1^B}^1 \theta_1^B [\bar{s}^B - \theta_1^B (1-\delta)] f(\delta|q) d\delta}_{\text{information blocking}}$$

Blocking interval: $[\delta^*, \bar{\delta}_{\text{inc}}]$ where $\bar{\delta}_{\text{inc}} \equiv \frac{v_{\text{inc}}}{(1-\kappa)(\bar{s}^E - \varphi_{\text{org}}^O)}$.

detail

Proposition 5.2 (Incumbent Blocking Restores Exploration)

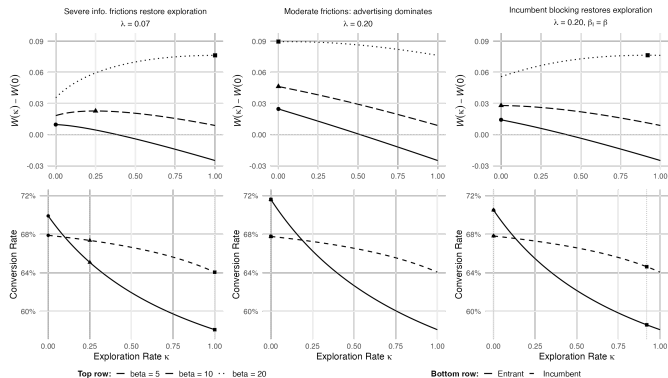
Suppose $\kappa_{\text{no-ads}}^* > 0$ and $v_{\text{inc}} > a^*$. Then $\kappa^* > 0$ whenever $\lambda < \bar{\lambda}'$, where

$$\bar{\lambda}' = \frac{\mathbb{E}_{\delta}[\Delta W(\delta)]}{\mathbb{E}_{\delta}[\Delta W(\delta) \cdot 1(\delta > \bar{\delta}_{\text{inc}})]} > \bar{\lambda}$$

If $\bar{\delta}_{\text{inc}} \geq 1$: $\bar{\lambda}' = \infty$, so $\kappa^* > 0$ for any λ .

- ▷ Blocking shrinks the set of entrants served by ads:
 $1(\delta > \bar{\delta}_{\text{inc}}) \subset 1(\delta \geq \delta^*)$
- ▷ For large J and β_I : $\bar{\delta}_{\text{inc}} \geq 1$
→ complete blocking
- ▷ $\bar{\lambda}' > \bar{\lambda}$: blocking makes exploration strictly easier to justify

Simulation: Advertising and Information Frictions



- **Left ($\lambda=0.07$):** even 7% participation reduces κ^* vs. no-ads
- **Center ($\lambda=0.20$):** $\kappa^* = 0$ for all β (Prop 4b)
- **Right:** + blocking restores $\kappa^* > 0$ at high β (Prop 5)

Connection to Empirical Findings

- Micro-found empirical results:

Empirical finding

High entrant conversion

Narrowing entry funnel

Ads raise entrant sales

Low ad adoption despite treatment

Model implication

⇒ Ranking-based screening; low κ

⇒ Platform maturation; incumbent saturation

⇒ Ads have signaling value

⇒ Incumbent blocking + information frictions

- Link stylized search equilibrium model with search-session-level data:
 - ▷ Market expansion test: controlling for consumer, keyword $\times$ date $\times$ number-of-search FE, Δ purchase prob $\rightarrow \Delta \mathcal{W}$ due to exploration and advertising
- Compared to existing models, we simplify the modeling of learning process but highlight:
 - ▷ Initial exploration decision (before dynamic accumulation of reviews/transactions)
 - ▷ Its relationship to advertising on an ad-supported platform

Conclusion

- First to document a **narrow(ing) entry funnel**
- Theory and evidence on entrants **(under-)exploration**
- Theory and evidence on **advertising's** (limited) ability to promote entry
- Next steps:
 - ▷ ads bidding: entrant constraints and incumbent strategic deterrence
 - ▷ “soft” explorations: prior-based ranking boosts
 - ▷ new products by incumbent vs. new sellers: bet on the horse vs. the jockey
 - ▷ entrant “learning-by-doing” and innovation

6 Appendix: Model

7 Appendix: Training

8 Appendix: Demand

A1. Platform-Mediated Search

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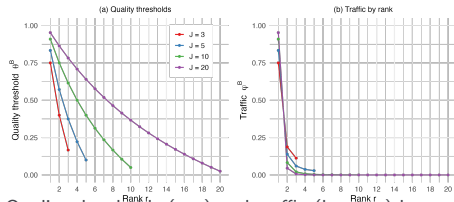
- $J \geq 1$ incumbent products, each with quality $\delta_j \in [0, 1]$ (price-inclusive, one-dimensional) (Athey and Ellison 2011; Vellodi 2018)
- Platform observes δ_j perfectly, ranks in descending order
- Consumer draws search cost $s_i \sim U[0, 1]$, inspects sequentially
- Stops upon match (prob δ_j , payoff 1 for both) or when $\theta^B(r, J) < s$

Equilibrium quality and traffic at rank r :

$$\theta^B(r, J) = \frac{J - r + 1}{J + r}$$

$$\varphi^B(r, J) = \bar{s}^B \cdot \prod_{r'=1}^{r-1} (1 - \theta^B(r', J))$$

Entry threshold: $\bar{s}^B = \theta^B(1, J) = J/(J+1)$



Quality thresholds (top) and traffic (bottom) by rank for $J = 5, 10, 20$. Larger $J \rightarrow$ steeper hierarchy, sharper traffic concentration at rank 1.

A2. Monotone Identity (Lemma A.1)

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Lemma A.1 (Monotone Identity)

For any monotone ranking $(\theta_1 \geq \theta_2 \geq \dots)$ with entry threshold $\bar{s}$:

$$W = \frac{1}{2\bar{s}} \sum_r \varphi_r \theta_r^2$$

Proof. Consumer with cost s inspects rank r whenever $\theta_r \geq s$. Let $p_r = \prod_{r' < r} (1 - \theta_{r'})$. Exchanging sum and integral:

$$W = \int_0^{\bar{s}} \sum_{\{r: \theta_r \geq s\}} p_r (\theta_r - s) ds = \sum_r p_r \int_0^{\theta_r} (\theta_r - s) ds = \sum_r p_r \cdot \frac{\theta_r^2}{2}$$

Defining $\varphi_r \equiv \bar{s} \cdot p_r$ yields the result. □ **Application:** Welfare in each regime is computed via the

Monotone Identity:

- Baseline: $W^B(J) = \frac{1}{2\bar{s}^B} \sum_r \varphi_r^B (\theta_r^B)^2$
- Organic: $W^O(q, J) = \frac{1}{2\bar{s}^B} \sum_{r=1}^{J+1} \varphi_r^O(q) (\theta_r^O(q))^2$, with $\theta_{r_{\text{org}}}^O = q$

A3. Three Regimes: Exploration

back to main

(i) *Exploration regime.* Entrant at rank 1, incumbents displaced one rank deeper. The entry threshold solves:

$$(q - \bar{s}^E) + \frac{1-q}{\bar{s}^B} \sum_{r=1}^{k^E} \varphi^B(r, J) (\theta^B(r, J) - \bar{s}^E) = 0$$

where $k^E = \max\{r : \theta^B(r, J) \geq \bar{s}^E\}$.

- First term: net value of inspecting entrant; sum: option value of deeper incumbents
- At $s = \bar{s}^B$: first term negative ($q < \bar{s}^B$), all summands $\leq 0 \Rightarrow$ surplus $< 0 \Rightarrow \bar{s}^E < \bar{s}^B$
- At $s = q$: first term vanishes, sum strictly positive ($\theta_1^B > q$) $\Rightarrow \bar{s}^E > q$
- Marginal consumer inspects $k^E \geq 1$ positions; exploration reduces entry but continuation value compensates
- Entrant's higher pass-through ($1 - q > 1 - \bar{s}^B$) channels more traffic to deeper ranks

Linearity: Per-consumer surplus $CS^E(\delta, s) = (\delta - s) + \frac{1-\delta}{\bar{s}^B} \sum_{r=1}^{k(s)} \varphi_r^B (\theta_r^B - s)$ is linear in δ for each s (match prob δ and pass-through $1-\delta$ both linear). Since $\bar{s}^E$ is independent of δ : $W^E(\delta, J) = A + B\delta \Rightarrow \mathbb{E}[W^E] = W^E(q, J)$

A3. Three Regimes: Organic and Full Information

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(ii) *Organic regime.* Entrant at rank $r_{\text{org}} = \lceil (J(1-q) + 1)/(1+q) \rceil$.

- When $q < \theta_1^B = J/(J+1)$: entrant below rank 1, deeper as q decreases or J increases
- Low- q entrant in large session: buried at position with negligible traffic $\rightarrow$ no learning possible
- Same linearity argument: $\mathbb{E}[W^O(\delta, J)] = W^O(q, J)$

(iii) *Full-information regime.* Entrant at true rank $r^*(\delta) = \min\{r : \theta^B(r, J) < \delta\}$.

- Both $r^*(\delta)$ and entry threshold $\bar{s}^{O*}(\delta) = \max(\delta, \bar{s}^B)$ depend on δ
- $W^*(\delta, J)$ is **not linear**: kinks at each $\theta^B(r, J)$ threshold where true rank improves
- When $\delta > \theta_1^B$: entry threshold rises to δ , drawing in additional consumers
- Heavy traffic skew toward top positions: correctly promoting high- δ entrant generates disproportionately more welfare gain than is lost by keeping low- δ at deep rank

Key implication: $W^*(\delta, J)$ convex $\Rightarrow \mathbb{E}[W^*] > W^*(q, J)$ (Jensen's inequality). This is the source of the learning gap $L > 0$.

A4. Proof of Proposition 1 (Diminishing Returns)

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Learning term derivation. Platform holds $\text{Beta}(2q, 2-2q)$ prior over δ , precision 2. Each consumer inspection generates a Bernoulli signal. Total entrant traffic: $T(\kappa) = \varphi_{\text{org}}^O + \kappa(\bar{s}^E - \varphi_{\text{org}}^O)$.

By Bernstein–von Mises, posterior variance $\propto 1/T(\kappa)$ (Fisher information scales linearly with traffic). Under quadratic loss, incremental learning value:

$$\Delta \mathcal{V}(\kappa) = C \left(\frac{1}{\varphi_{\text{org}}^O} - \frac{1}{T(\kappa)} \right) = \frac{C \kappa (\bar{s}^E - \varphi_{\text{org}}^O)}{\varphi_{\text{org}}^O T(\kappa)}$$

Normalizing so first explored consumer delivers $\beta \cdot L$: $C = \frac{\beta \cdot L \cdot (\varphi_{\text{org}}^O)^2}{\bar{s}^E - \varphi_{\text{org}}^O}$. The $(\bar{s}^E - \varphi_{\text{org}}^O)$ terms cancel:

$$\Delta \mathcal{V}(\kappa) = \frac{\beta \cdot L \cdot \varphi_{\text{org}}^O \cdot \kappa}{T(\kappa)} = \text{MLB}(\kappa) \cdot \kappa$$

Existence and uniqueness. $\widetilde{\text{MLB}}(\kappa) \equiv \frac{d}{d\kappa} [\text{MLB}(\kappa) \cdot \kappa] = \beta \cdot L \cdot (\varphi_{\text{org}}^O)^2 / T(\kappa)^2$ is:

(i) $\widetilde{\text{MLB}}(0) = \beta \cdot L > 0$; (ii) strictly decreasing; (iii) $\widetilde{\text{MLB}}(1) < \text{MLC}$ for $q < \bar{s}^B$. Define $g(\kappa) = \widetilde{\text{MLB}}(\kappa) - \text{MLC}$. By IVT: unique $\kappa^* \in (0, 1)$ with $g(\kappa^*) = 0$. When $\widetilde{\text{MLB}}(1) \geq \text{MLC}$: $\kappa^* = 1$.



A5. Per-Session Conversion Rate and Proposition 2 (Proof)

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Lemma A.2 (Per-Session Entrant Conversion Rate)

For any session (q, J) and any $\kappa \in [0, 1]$: $\frac{\mathbb{E}_\delta[\text{entrant needs met}]}{\mathbb{E}_\delta[\text{entrant traffic}]} = q$

Proof. Under exploration, entrant traffic $= \bar{s}^E$; under organic, traffic $= \varphi_{\text{org}}^O$. Both independent of δ . Combined traffic $T = \kappa \bar{s}^E + (1-\kappa) \varphi_{\text{org}}^O$. Needs met $= \delta \cdot T$. Since $T \perp \delta$: ratio $= \mathbb{E}[\delta] = q$. $\square$

Proof of Proposition 2 (Exploration Dilutes Baseline Screening).

Aggregate entrant conversion: $\bar{C}_E^J(\kappa) = \frac{\sum_q q \cdot T(q, J, \kappa)}{\sum_q T(q, J, \kappa)}$, where $T(q, J, \kappa) = \kappa \bar{s}^E + (1-\kappa) \varphi_{\text{org}}^O$.

For $\kappa' > \kappa$, the traffic ratio $T(q, \kappa')/T(q, \kappa)$ is decreasing in φ_{org}^O :

$$\frac{\partial}{\partial \varphi_{\text{org}}^O} \frac{T(q, \kappa')}{T(q, \kappa)} = \frac{\bar{s}^E \cdot (\kappa - \kappa')}{[\kappa \bar{s}^E + (1 - \kappa) \varphi_{\text{org}}^O]^2} < 0$$

Since φ_{org}^O is weakly increasing in q , the likelihood ratio is decreasing in q . By MLRP, κ' shifts traffic weight toward low- q sessions. Since q is the quantity averaged: $\bar{C}_E^J(\kappa') < \bar{C}_E^J(\kappa)$. $\square$

A6. Per-Session Comparative Statics (Lemma A.3)

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Lemma A.3 (Per-Session Comparative Statics)

For each session (J, q) with $q < \bar{s}^B$: $\frac{\partial \kappa^*}{\partial J} < 0$ and $\frac{\partial \kappa^*}{\partial q} > 0$

Both follow by implicit differentiation of $\widetilde{\text{MLB}}(\kappa^*) = \text{MLC}(q)$.

$\partial \kappa^* / \partial J < 0$:

- Denominator $\partial \widetilde{\text{MLB}} / \partial \kappa < 0$ (strictly decreasing)
- $\partial \text{MLC} / \partial J > 0$: more incumbents raise welfare more under organic ranking than under exploration
- $\partial \widetilde{\text{MLB}} / \partial J \leq 0$: information gain from correct ranking shrinks as incumbent quality rises
- Numerator positive + negative denominator $\Rightarrow \partial \kappa^* / \partial J < 0$

$\partial \kappa^* / \partial q > 0$:

- MLC channel: $\partial \text{MLC} / \partial q < 0$ (higher q narrows exploration–organic welfare gap)
- MLB channel: ambiguous (higher q raises φ_{org}^O but may reduce L)
- Extensive margin at $\kappa = 0$: $\beta \cdot L$ does not depend on φ_{org}^O
- Result holds when MLC reduction dominates—verified numerically across full parameter space

A6 (cont). Proof of Proposition 3 (Optimal Exploration)

[back to main](#)

Aggregate optimum: κ^* maximizes $\mathcal{W}(\kappa) = \sum_{J,q} w(J, q) \cdot W(\kappa; J, q)$.

$\partial \kappa^* / \partial \beta \geq 0$: $\text{MLB}(\kappa) \cdot \kappa = \beta \cdot L \cdot \varphi_{\text{org}}^O \cdot \kappa / T(\kappa)$ is increasing in β for every $\kappa > 0$. By monotone comparative statics (Milgrom and Shannon 1994)

, κ^* is weakly increasing in β .

$\partial \kappa^* / \partial \alpha \geq 0$: Higher α shifts Pareto weight toward smaller J . By Lemma A.3, smaller sessions have higher optimal exploration. Reweighting increases aggregate marginal benefit at every $\kappa > 0$.

$\partial \kappa^* / \partial \mu \geq 0$: Higher μ shifts quality distribution upward. By Lemma A.3, per-session κ^* is increasing in q . Higher q reduces MLC in every session, increasing aggregate marginal benefit.

Interpretation: As platform matures (α falls, J increases): MLC rises, κ^* declines—consistent with the narrowing entry funnel documented in Section 2.

A7. Learning Rate Thresholds

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Interior exploration requires $\beta \in (\beta_{\text{low}}, \beta_{\text{high}})$:

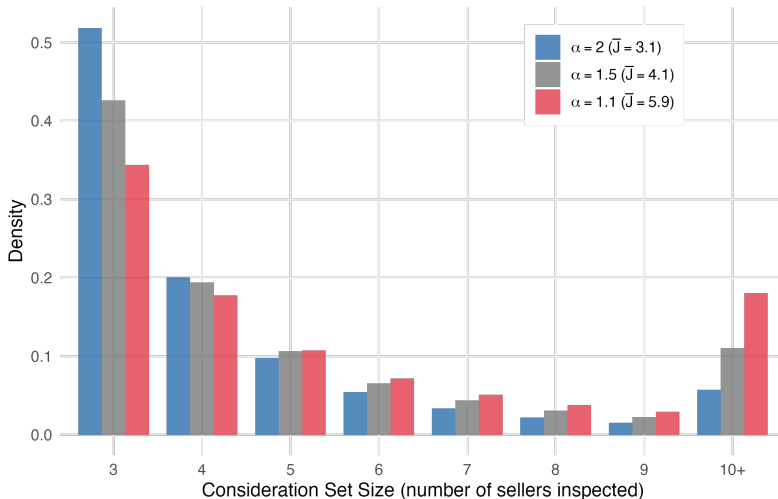
$$\beta_{\text{low}} = \frac{\sum_{J,q} w \cdot \text{MLC}}{\sum_{J,q} w \cdot L}, \quad \beta_{\text{high}} = \frac{\sum_{J,q} w \cdot \text{MLC}}{\sum_{J,q} w \cdot L \cdot (\varphi_{\text{org}}^O / \bar{S}^E)^2}$$

μ	$\alpha = 2.5$ ($\bar{J} = 2.7$)	$\alpha = 2.0$ ($\bar{J} = 3.1$)	$\alpha = 1.5$ ($\bar{J} = 4.1$)	$\alpha = 1.2$ ($\bar{J} = 5.3$)	$\alpha = 1.1$ ($\bar{J} = 5.9$)
0.10	(25, 1016)	(28, 1189)	(34, 1502)	(41, 1831)	(44, 1982)
0.25	(7, 181)	(8, 213)	(10, 272)	(12, 335)	(12, 363)
0.50	(2.4, 12)	(2.7, 14)	(3.3, 16)	(3.8, 19)	(4.1, 21)
0.75	(0.4, 1.5)	(0.5, 1.9)	(0.7, 2.7)	(0.9, 3.5)	(1.0, 3.9)

- Average entrant quality μ is quantitatively dominant: at $\alpha = 1.5$, a platform with $\mu = 0.10$ requires $\beta > 34$ to begin exploring; $\mu = 0.75$ begins at $\beta > 0.7$
- β_{high} typically 1–2 orders of magnitude larger than $\beta_{\text{low}} \Rightarrow$ full exploration almost never optimal; interior regime dominates

A8. Model-Predicted Consideration Set Sizes

[back to main](#)



- Model-predicted density of consideration set sizes from baseline equilibrium, aggregated over $J \in \{2, \dots, 100\}$ with truncated Pareto weights at three concentration levels α

A9. Sign Characterization of ΔW (Proof of Prop 4)

[back to main](#)

Fix (J, q) with $q < \theta_1^B$. $\Delta W(\delta) \equiv \beta[W^*(\delta, J) - W^O(q, J)] - [W^O(\delta, J) - W^E(\delta, J)]$.

Since $W^*(\delta, J)$ piecewise linear & convex, $W^O(\delta, J) - W^E(\delta, J)$ linear $\Rightarrow \Delta W$ convex. FOC: $\Delta W(\delta^*) = 0$.

$\delta > \delta^*$: $\Delta W'(\delta^*) = \beta W^{*'}(\delta^*) - [W^{O'} - W^{E'}] > 0$. Convexity $\Rightarrow \Delta W'$ non-decreasing, so $\Delta W > 0$ on $(\delta^*, 1]$.

$\delta \in [q, \delta^*)$: At $\delta = q$: $W^*(q, J) = W^O(q, J)$ while $\text{MLC} > 0 \Rightarrow \Delta W(q) = -\text{MLC} < 0$. Convexity on $[q, \delta^*]$:

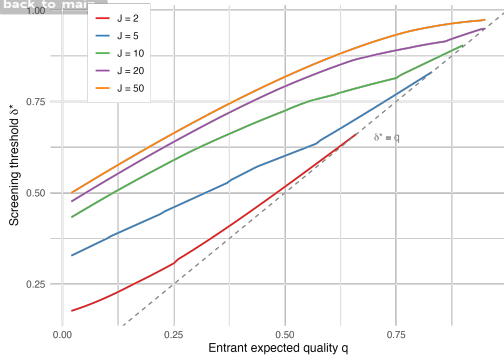
$$\Delta W(\delta) \leq \frac{\delta^* - \delta}{\delta^* - q} \Delta W(q) + \frac{\delta - q}{\delta^* - q} \Delta W(\delta^*) = \frac{\delta^* - \delta}{\delta^* - q} (-\text{MLC}) < 0$$

$\delta \in [0, q)$: At $\delta = 0$: $W^*(0, J) = W^B(J) \leq W^O(q, J)$ and $W^O(0, J) \geq W^E(0, J) \Rightarrow \Delta W(0) \leq 0$. Convexity gives $\Delta W(\delta) \leq \frac{\delta}{q} (-\text{MLC}) < 0$ for $\delta > 0$. □

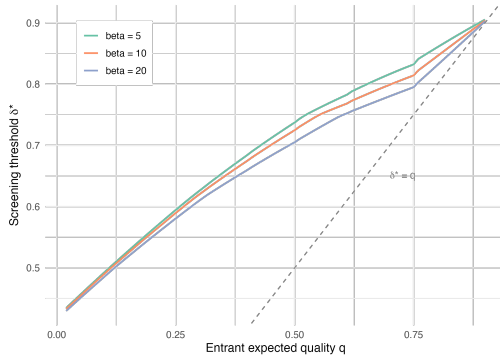
Frictionless $\kappa^* = 0$: All $\delta \geq \delta^*$ self-select into paid channel; residual pool has $\Delta W < 0$ everywhere. Exploration of any residual type is welfare-reducing.

A10. Welfare-Maximizing Screening Threshold δ^*

[back to main](#)



(a) Varying J ($\beta = 10$)



(b) Varying β ($J = 10$)

Analytical characterization: $\delta^* = q + \frac{\text{MLC}(q)}{\beta \cdot W^{*'} + |\text{MLC}'|}$

- Numerator: displacement cost at the prior
- Denominator: rate at which net benefit grows with δ (marginal learning + MLC reduction)
- Kinks at incumbent thresholds $\theta^B(r, J)$ where $W^{*'}$ increases discretely
- Larger J raises δ^* (displacement cost $\uparrow$): higher β lowers δ^* (learning value $\uparrow$)

A11. Information Frictions: Why $\kappa^* > 0$

[back to main](#)

Fraction $\lambda \in [0, 1]$ participates in advertising. Among participants: only $\delta \geq \delta^*$ promoted; all others fall back to κ mechanism.

At $\kappa = 0$, marginal welfare from raising κ :

$$\left. \frac{dW}{d\kappa} \right|_{\kappa=0} = \underbrace{(1-\lambda) \mathbb{E}_{\delta}[\Delta W(\delta)]}_{\text{non-participants}} + \underbrace{\lambda \mathbb{E}_{\delta}[\Delta W(\delta) \cdot 1(\delta < \delta^*)]}_{\text{sub-threshold participants}}$$

Write $A = \mathbb{E}[\Delta W \cdot 1(\delta < \delta^*)] < 0$ and $B = \mathbb{E}[\Delta W \cdot 1(\delta \geq \delta^*)] > 0$.

Then $dW/d\kappa|_{\kappa=0} = A + (1-\lambda)B > 0$ iff $\lambda < \frac{A+B}{B} = \bar{\lambda}$.

- Since $A < 0$ and $A + B > 0$ (by $\kappa_{\text{no-ads}}^* > 0$): $\bar{\lambda} \in (0, 1)$
- $\bar{\lambda}$ close to 1 when sub-threshold loss $|A|$ small relative to above-threshold benefit B
- $\bar{\lambda}$ approaches 0 when sub-threshold losses are large

A12. Incumbent Blocking: v_{inc} and Complete Deterrence

[back to main](#)

Second-price position auction. Incumbent's total value of retaining rank 1:

$$v_{\text{inc}}(q, J) = \underbrace{\theta_1^B [\bar{s}^B - \bar{s}^E (1-q)]}_{\text{static: revenue loss from dropping to rank 2}} + \underbrace{\beta_I \int_{\theta_1^B}^1 \theta_1^B [\bar{s}^B - \theta_1^B (1-\delta)] f(\delta|q) d\delta}_{\text{dynamic: information blocking}}$$

Static term: at rank 2, incumbent receives traffic $\bar{s}^E (1-q)$ instead of $\bar{s}^B$.

Dynamic term: if platform observes $\delta \geq \theta_1^B$, it permanently promotes entrant. β_I analogous to β .

Entrant WTP (liquidity-constrained): $v(\delta) = (1-\kappa)(\bar{s}^E - \varphi_{\text{org}}^O) \cdot \delta$

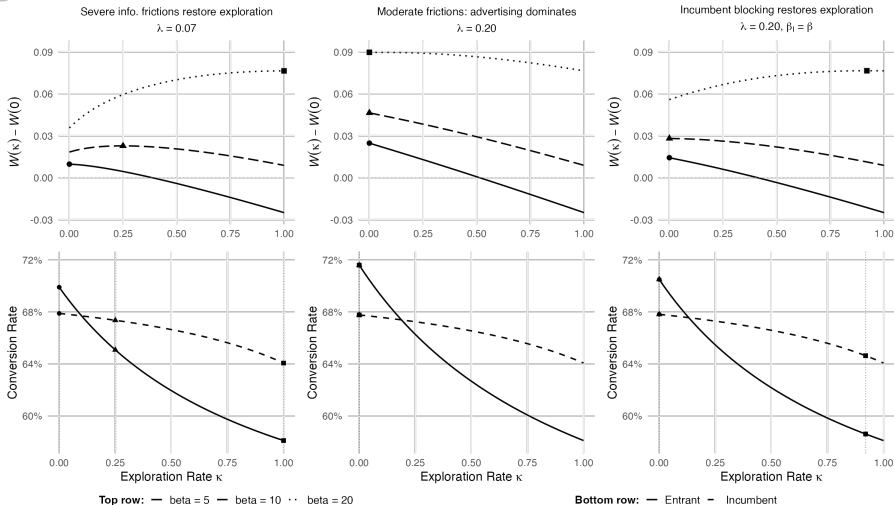
Blocking interval: $[\delta^*, \bar{\delta}_{\text{inc}}]$ where $\bar{\delta}_{\text{inc}} = v_{\text{inc}} / [(1-\kappa)(\bar{s}^E - \varphi_{\text{org}}^O)]$

Complete blocking ($\bar{\delta}_{\text{inc}} \geq 1$): no participant can outbid incumbent, so all entrants are in same position as without advertising $\Rightarrow \kappa^* > 0$ follows from $\kappa_{\text{no-ads}}^* > 0$ for any λ .

Partial blocking ($\delta^* < \bar{\delta}_{\text{inc}} < 1$): relaxes $\bar{\lambda}$ threshold but does not eliminate it.

A13. Dynamic Entrant Bidding (Robustness)

[back to main](#)



- Relaxing liquidity constraint: entrant with $\delta > \theta_1^B$ anticipates permanent promotion, bids $v^{\text{dyn}}(\delta) = (1-\kappa)(\bar{s}^E - \varphi_{\text{org}}^O)\delta + \beta^E \delta(\delta - \varphi_{\text{org}}^O)$

6 Appendix: Model

7 Appendix: Training

8 Appendix: Demand

Entrepreneur Training

back

Progress-tracking
Module

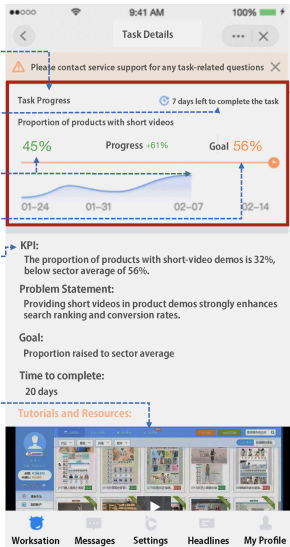
Time Left

Current KPI Level

Target KPI Level

KPI Definition

Tutorials



Task-based entrepreneur training flowchart

- ▶ Quantitative: focus on strategy and performance KPIs
- ▶ Customized: an AI algorithm dynamically matches sellers to training tasks based on KPIs
- ▶ Benchmarked: KPI tracked against sector average
- ▶ Independent: no direct impact on search ranking & recommendation

Types of Tasks

Task (Examples)	KPI	Area
Complete an order Offer product-level discounts Setup seller-wide discounts Setup 7-day free return	payment received using product discount tool on 1+ products use of “red-pocket” (coupons) 7-day free return status	basic basic basic basic

1. basic operations: complete listings; offer pricing and discounts
2. marketing: conduct SEO (e.g. short-video description) and paid ads
3. ratings: improve seller and product reviews
4. customer service: improve service efficiency (e.g. chatbot adoption)

Types of Tasks

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Complete an order Offer product-level discounts Setup seller-wide discounts Setup 7-day free return	payment received using product discount tool on 1+ products use of “red-pocket” (coupons) 7-day free return status	basic basic basic basic
Optimize product titles set up ad account Set up automated ad campaign	use of product title-setting module deposit funds into ads account conducted keywords campaign and budget	marketing marketing marketing

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Optimize product titles set up ad account Set up automated ad campaign	use of product title-setting module deposit funds into ads account conducted keywords campaign and budget	marketing marketing marketing
Tips to improve delivery ratings Tips to improve service ratings Offer targeted coupons during customer inquiries Set up chatbot automation in customer chat app	logistics component of seller rating service component of seller rating used discount tools in customer chat app completed basic setup of store information	ratings ratings/service service/basic service

1. basic operations: complete listings; offer pricing and discounts
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Treated New Sellers Earn Higher Revenues

We reached 2+ mil new sellers in one year and got significant ITT/TOT on traffic and revenue.



Prominent Front-page Module

Task #1: Unclaimed

Current KPI Level

Target KPI Level

of Stores Claiming Task

Experiment Design rollout

- ▶ $\frac{1}{12}$ potential entrants were randomized into treatment when they registered (May 2019; later raised to $\frac{1}{6}$ and then 45%)
- ▶ offered at no cost to treatment-group sellers
- ▶ key limitation: no task-level or algo randomization

Takeup details

- ▶ 24% opted in, 13% completed 1+ tasks
- ▶ higher among entrants that were male-led, in southern provinces, in the 60-90th revenue pctl
- ▶ 77% of task completion was within the first three months

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Treated New Sellers Earn Higher Revenues

We reached 2+ mil new sellers in one year and got significant ITT/TOT on traffic and revenue.



Results:

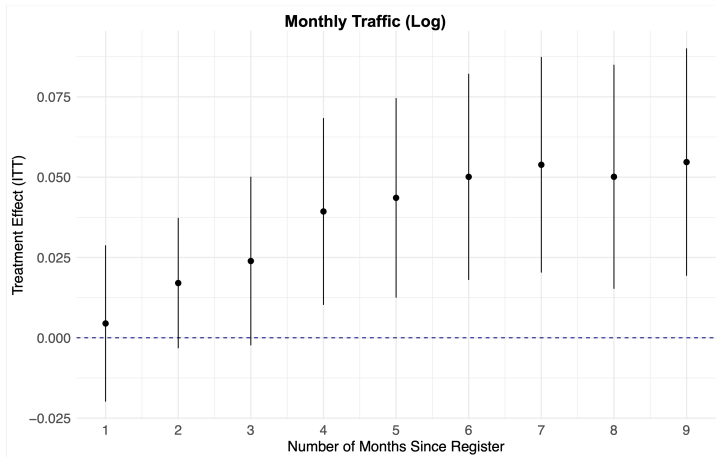
	ITT (Potential Entrants)	ITT (Entrants)	TOT
Traffic	+1.3%	+4.1%	+5.2%
Revenue	+1.7%	+5.3%	+6.6%
Has Revenue	+0.8% [†]	+1.5%	+3.2% [†]

Notes: [†] denotes $p < 0.05$, otherwise $p < 0.01$; unit: seller-month.

- ▷ no detectable impact on conversion, ratings, and returns
- ▷ may not indicate lack of quality impact:
 - * more traffic → more & *different* consumers
 - * noisier metrics

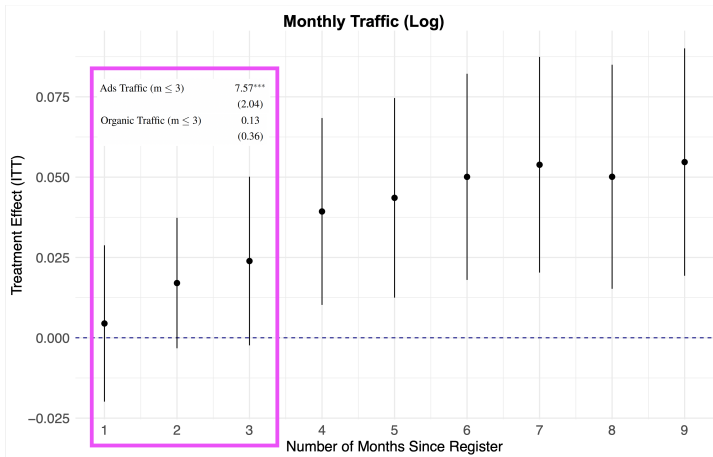
itt ittpe att ittdyn ittpedyn itthet ittquant ittqual

Treatment Effect Dynamics (Entrants)



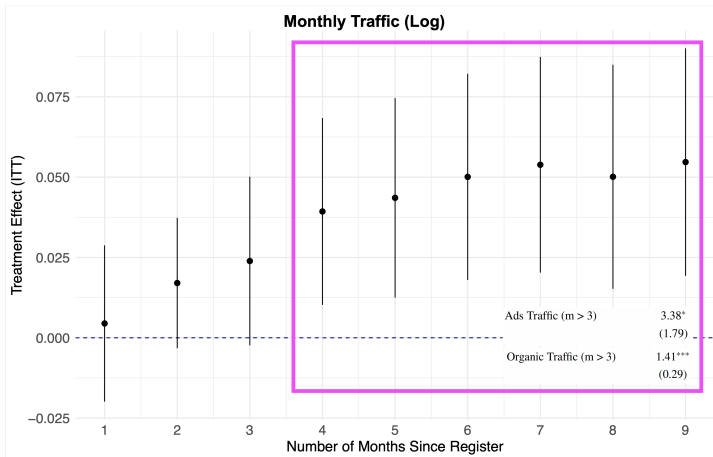
- Follow May - Aug 2019 entrant cohort for 9 months
- Steady increase in traffic, stabilizing around month 6
- paid traffic increased first and sharply; timing coincided with training intensity ($t \leq 3$)
- organic traffic increased later (post “cold-start”); smaller % change but bigger contribution to total traffic

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Seller Strategies - What Did They Do Differently?

[demo](#)[back](#)

Variable	Treatment (%)	Dep Var Mean
Marketing - Paid		
# Ads Purchased	2.96*** (1.00)	1.57
Marketing - SEO		
Has Video Demo ($t \leq 3$)	1.97*** (0.72)	0.11
Has Title Change	-0.02 (0.62) (0.02)	0.05
Ratings		
% of Good Reviews	0.00 (0.02)	0.99
Star Rating	0.35 (0.36)	3.77
Pricing		
Avg. Price Per Buyer	0.50 (0.59)	80.72
Avg. Price Per Product	0.03 (0.59)	46.22
Service		
Active Time (min)	-0.41 (0.67)	19.17

- Training worked by nudging sellers to advertise
 - ▷ Treated sellers did post more short-video product descriptions, but we saw no contemporary lift in organic traffic.
- No such effect (or information friction) exists on pricing, positioning, etc. [retention](#)
 - ▷ despite similar coverage and higher task take-up and completion rates
 - ▷ does not suggest “optimal” pricing

6 Appendix: Model

7 Appendix: Training

8 Appendix: Demand

Demand Estimation

- Consumer Demand: [details](#) [estimates](#)
 - ▷ account for rich consumer preference heterogeneity
 - ▷ instrument for prices and product variety
IV: platform's rule enforcement + treatment assignment
 - ▷ estimate seller-specific unobserved "ability" ξ_j [details](#)

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Statistic	Mean	St. Dev.	Min	Pctl(25)	Pctl(75)	Max
Control	0.110	0.591	−1.843	−0.087	−0.016	6.024
Treatment	0.113	0.589	−1.789	−0.088	−0.013	5.256
Incumbent	0.096	0.577	− 3.253	−0.108	−0.008	6.524

Demand Estimation

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 - ▷ instrument for prices and product variety
IV: platform's rule enforcement + treatment assignment
 - ▷ estimate seller-specific unobserved “ability” ξ_j [details](#)
- Platform matching [details](#)
 - ▷ account for path-dependence and growth compounding in platform traffic allocation

Model: Consumer Demand [back](#)

Consumer i (searching a keyword) visits a set of sellers $j \in K_i$ and chooses

$$\max_{j \in K_i} U_{ij} = V_j + \epsilon_{ij} = x_j \beta - \alpha p_j + \xi_j + \epsilon_{ij} \quad (1)$$

- p_j : average price level of seller j (observed)
- x_j : seller j 's strategies and characteristics (observed)
- ξ_j : seller j 's post-treatment quality (*unobserved*) → **main interests**

Assuming logit error for ϵ_{ij} and normalize outside option of not to purchase to zero, we get

$$P_{ij} = \frac{\exp(x_j \beta - \alpha p_j + \xi_j)}{1 + \sum_{k \in K_i} \exp(x_k \beta - \alpha p_k + \xi_k)}$$

- P_{ij} : consumer i 's likelihood to purchase from seller j
- Instrument for pricing and strategies: [details](#)
- Consumer welfare and seller's revenues: [details](#)
- Estimated with simulated maximum likelihood on the consumer-seller-search session sample

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Estimated Results: Sellers' Underlying Quality ξ_j [back](#)

- Underlying quality $\sim$ demand residuals
 $\Rightarrow$ sellers with purchase records are different
- Both treated and control new sellers have higher underlying quality (ξ_j) than incumbents

[dist](#)

	<i>Dependent variable:</i>		
	Full	ξ_j Purchase	No Purchase
	(1)	(2)	(3)
Control	0.015 (0.010)	0.142*** (0.032)	0.021*** (0.003)
Treatment	0.017* (0.019)	0.109*** (0.055)	0.013** (0.005)
Constant	0.096*** (0.003)	0.615*** (0.007)	-0.110*** (0.001)
Observations	52,241	14,624	37,617

Estimated Results: Sellers' Underlying Quality ξ_j [back](#)

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Observations	52,241	14,624	37,617

Model: Consumer-Seller Matching [back](#)

Number of visitors sellers attract determined by past observables

$$\mathcal{T}_{jt} = f(\mathcal{T}_{jt-1}, C_{jt-1}, C_{jt-2})$$

$\mathcal{T}_{jt}$ number of visitors, C_{jt} conversion rate for seller j in period t

- Approximate how the platform match consumers to sellers based on key observables
- Training affects the matching through $\mathcal{T}_{jt-1}$
- Underlying quality ξ_j affects traffic assignment only through conversion rates C_{jt-k}
- Estimate with new sellers' monthly panel

Model: Endogenous Strategies [back](#) [model](#)

We do not explicitly model sellers' strategies - instead, we use **instruments** to solve for endogenous choices of sellers' strategies

$$\begin{bmatrix} p_{jt} \\ \text{prod}_{jt} \end{bmatrix} = Z^{iv} \beta^{fs} + \xi_j^{fs} + \xi_t^{fs} + \xi_m^{fs} + \epsilon_{jt}$$

- Correlation between ξ_j^{fs} and ξ_j^{conv} $\rightarrow$ selection on strategy
- Z^{iv} : platform's stringency of regulation enforcement (product category - temporal variations) and training assignment
- Include calendar time ξ_t^{fs} and main industry ξ_m^{fs} fixed effect

Consumer Welfare and Revenues [back](#)

Consumer Welfare

- Use standard logsum formula with logit specification

$$CS = \frac{1}{\alpha} \sum_i \log \left[\sum_{j \in K_i} \exp(V_j) \right]$$

Revenues

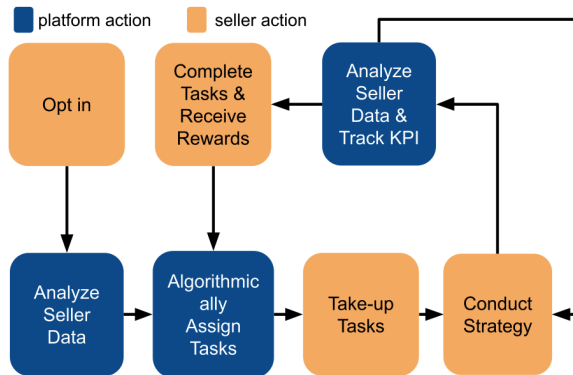
- Consumer's choice probability determines seller j 's exposure and revenue

$$R_j = \sum_i \frac{\exp(V_j)}{1 + \sum_{k \in K_i} \exp(V_k)} p_j$$

- Only relevant when seller j actually appears in consumer i 's set

Estimation: simulated maximum likelihood on the consumer-seller matched pair sample

Training: AI-Facilitated Customization back

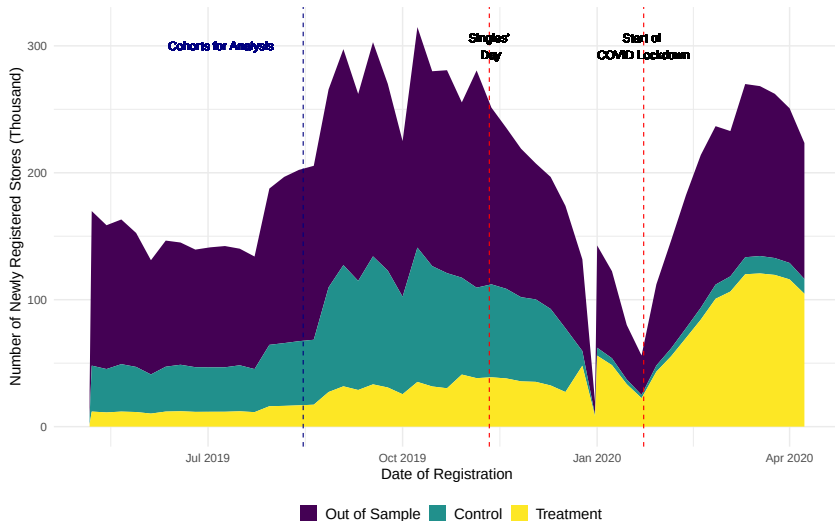


Development

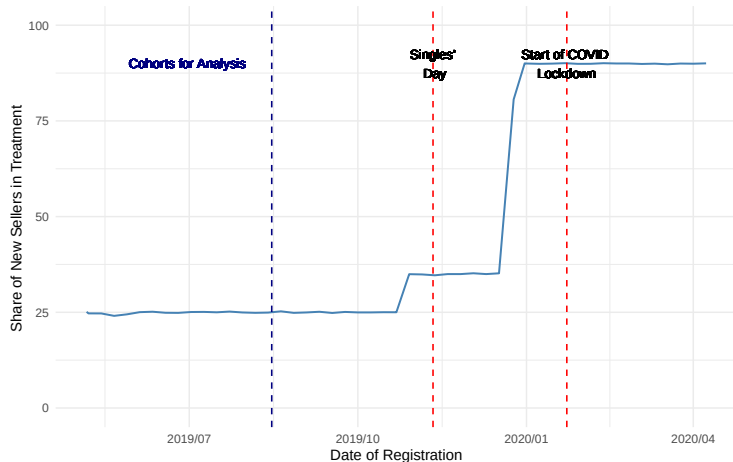
- ▷ map *pre-existing* training materials to KPI thresholds
 - * action tasks: e.g. set up an ad campaign (keyword bidding)
 - * outcome tasks: e.g. improve conversion of inquiring customers
- ▷ sequence tasks w/ a proprietary algo
 - * historical correlation with next-quarter revenue growth + manual adjustment
 - * low-tech: no causal evidence or online re-training
- ▷ construct UI with progress tracking and benchmark
- ▷ deliver tasks based on each participant's KPI

Training: Roll-Out [back](#)

Figure: Experimental Sample and Treatment Assignment Over Time

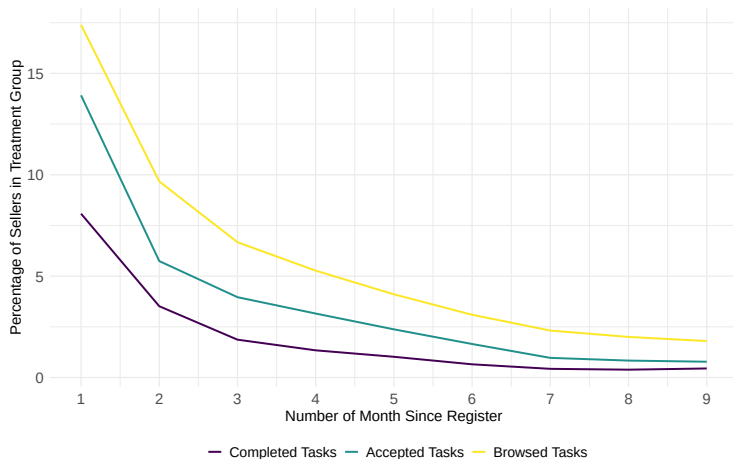


Experimental Design and Implementation [back](#)



- **randomize** new sellers' **access** to training
 - ▷ **26%** of new sellers in the main study cohorts received access
 - ▷ Reached 2+ million new sellers since launch in 5/2019

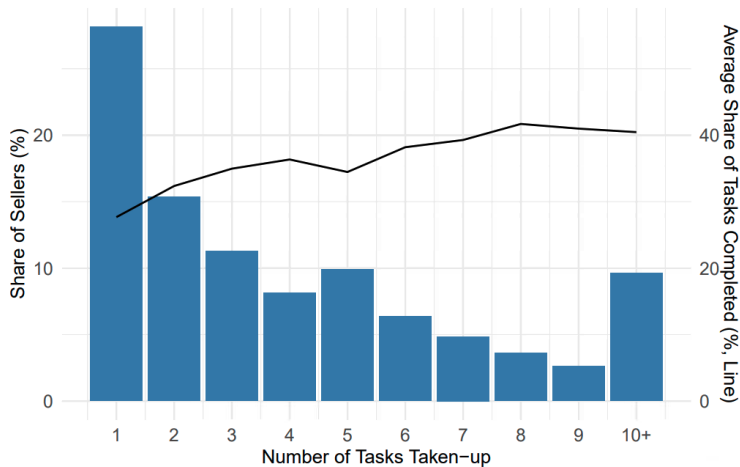
Experimental Design and Implementation [back](#)



- **Take-up is 24%** [tasks](#)
 - ▷ 50% of tasks taken-up in the first month
 - ▷ 62% of task-takers completed at least one task
 - ▷ Higher take-up among registered firms, men, coastal regions, agile [select](#)

Training Participation: Task Take-up and Completion [back](#)

Big variations in training participation: 10% of sellers are intensely involved and took up 10+ tasks



Characteristics of Participants [back](#)

	Participants	Non Participants	P - NP
Is Firm	0.275 (0.446)	0.266 (0.442)	0.008*** (3.18)
Female Owner (among individual sellers)	0.442 (0.497)	0.458 (0.497)	-0.16*** (1.27)
Region: Coastal South	0.518 (0.500)	0.413 (0.492)	0.105*** (3.573)
Region: West	0.089 (0.284)	0.126 (0.332)	-0.038*** (21.41)
List Products on 1st Day	0.245 (0.430)	0.204 (0.403)	0.04*** (16.04)
Observations	36,189	140,837	177,026

Selection on Take-Up [back](#)

OLS estimates on training take-up:

- Training participants are much more active than average sellers on almost all main performance metrics

	<i>Dependent variable:</i>							
	Log Revenues (1)	Any Revenues (2)	Revenues (3)	Log # Visitors (4) (5)		Log # Buyers (6) (7)		Conversion (8)
Take-up Tasks	1.847*** (0.017)	0.223** (0.002)	0.771*** (0.019)	1.735** (0.014)	0.703*** (0.015)	0.825*** (0.009)	0.686*** (0.012)	0.005*** (0.0004)
Dep Var Mean	-	0.19	-	-	-	0.57	1.4	0.04
Sample	Full	Full	Earn Revenues	Full	Have Visitors	Full	Have Visitors	Have Visitors
Observations	1,593,234	1,593,234	314,376	1,593,234	646,894	1,593,234	646,894	646,894
R ²	0.193	0.204	0.108	0.274	0.128	0.160	0.098	0.048
Adjusted R ²	0.193	0.204	0.1087	0.274	0.128	0.159	0.097	0.047

Notes: Standard errors clustered by seller. Revenues, number of visitors and number of buyers measured in log.

Training ITT (Entrants) [back](#)

Table: Intend-to-Treat Effects on Seller Outcome Among Sellers with Positive Traffic

	<i>Dependent variable:</i>			
	Log Revenues (3)	Log Traffic (5)	Log Customers (7)	Conversion Rate (8)
Treatment	0.026*** (0.010)	0.024*** (0.008)	0.018*** (0.006)	0.0001 (0.0002)
Dep Var Mean	2.32	4.29	1.40	0.04
Adjusted R ²	0.081	0.111	0.070	0.043

Notes: This table replicates Table 11 but we instead use a subsample of sellers that have seen positive traffic in at least one month during our research period, $N = 2,593,762$. Conversion rate is defined as the share of visitors (traffic) that make a purchase within a week.

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Table: Intend-to-Treat Effects on Seller Outcome Among All Registered Sellers

	<i>Dependent variable:</i>			
	Log Revenues (1)	Has Revenues (2)	Log Traffic (3)	Log Customers (4)
Treatment	0.017*** (0.006)	0.002** (0.001)	0.013** (0.006)	0.008*** (0.003)
Dep Var Mean	1.39	0.19	1.73	0.57
Adjusted R ²	0.132	0.152	0.207	0.105

Notes: $N = 6,409,062$. Dependent variables are summed to the month level. Logged values are taken after adding one to the original levels. “Has revenues” is an indicator for earning positive revenues in the month. All regressions include cohort, month, and sector fixed effect as specified by equation ???. Dependent variable means are calculated with sellers in the control group across all months. Standard errors are clustered on the seller level. ** significant at 5% level and *** significant at 1% level.

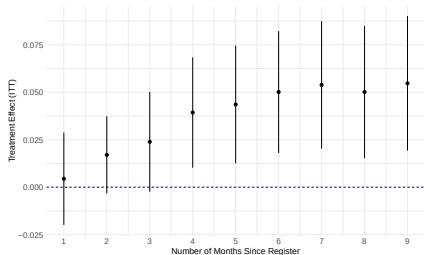
Table: ATT Results on Main Outcomes: 2SLS Estimates with Training Take-up

	<i>Dependent variable:</i>					
	Takes up Training (1)	Has Revenues (2)	Revenues (3)	# Visitors (4)	# Buyers (5)	Conversion Rate
Treatment	0.257*** (0.001)					
Take-up Training		0.006** (0.003)	0.066*** (0.024)	0.052** (0.021)	0.032*** (0.012)	0.0001 (0.001)
Dep Var Mean	0	0.19	1.39	1.73	0.57	0.04
Observations	6,409,062	6,409,062	6,409,062	6,409,062	6,409,062	2,593,762
R ²	0.232	0.153	0.133	0.208	0.106	0.049

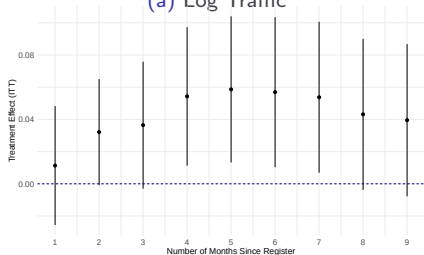
Notes: This table reports the 2SLS (two-stage-least-square) estimates for the training ATT (average treatment effects on the treated). Column 1 reports the first-stage estimates, in which we regress the take-up indicator (for whether a treatment-group seller has taken up any task) on the treatment indicator. Columns 2 to 6 report the 2SLS estimates. All regressions include cohort, month, and sector fixed effects, with standard errors clustered at the seller level. Dependent variable means are calculated among control-group sellers. ** significant at 5% level and *** at 1%.

Training ITT Dynamics (Entrants)

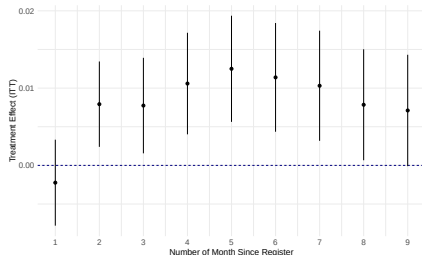
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(a) Log Traffic



(c) Log Revenues

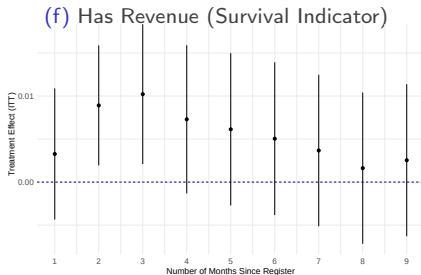
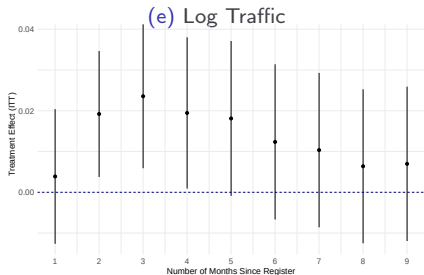
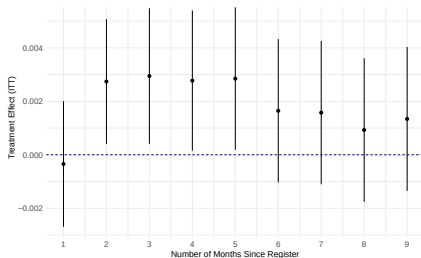
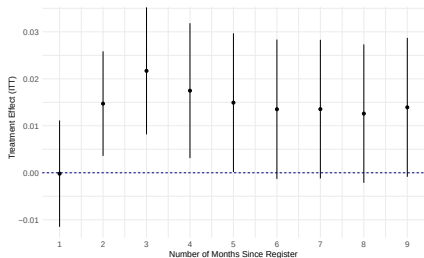


(b) Has Revenue (Survival Indicator)



(d) Log Customers

Training ITT Dynamics (Potential Entrants)

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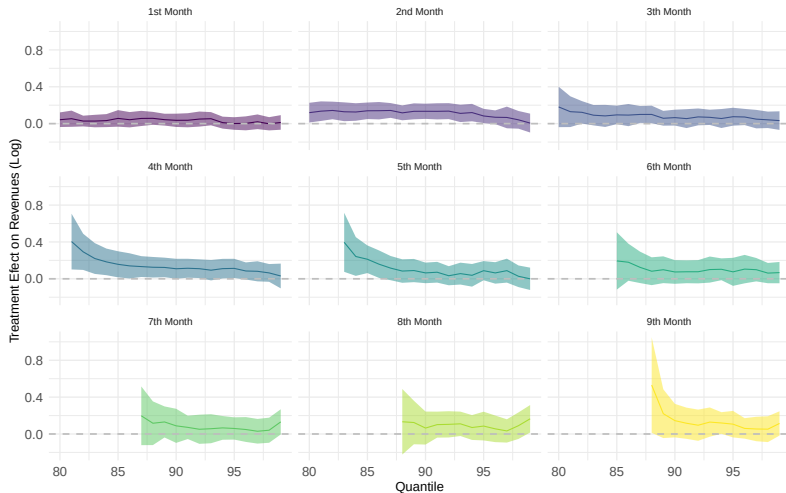
(g) Log Revenues

(h) Log Customers

Training ITT Quantile Effect

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No significant difference on treatment effect by size quantile



Training ITT: Ratings and Returns [back](#)

Training has **no** significant ITT on ratings and returns

measured with sellers' ratings (product description, logistic and service), consumers' refunds and complaints, or likelihood of rule violations

Variable	Treatment	Dep Var Mean	Variable	Treatment	Dep Var Mean
Ratings			Refunds and Reviews		
Products	0.004 (0.004)	1.25	% Refund (Amount)	-0.0004 (0.001)	0.21
Service	0.004 (0.004)	1.26	% Complaints	-0.003 (0.006)	0.05
Logistics	0.005 (0.004)	1.26	Rule Violations	0.0002 (0.001)	0.22
			% Good Reviews	-0.00001 (0.0002)	0.99

Notes: Table presented estimated coefficients β on treatment assignment dummy with specification $??$. Standard errors clustered by seller.

Demand Estimation

- Consumer Demand: [details](#) [estimates](#)
 - ▷ account for rich consumer preference heterogeneity
 - ▷ instrument for prices and product variety
IV: platform's rule enforcement + treatment assignment
 - ▷ estimate seller-specific unobserved "ability" ξ_j [details](#)

Demand Estimation

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Statistic	Mean	St. Dev.	Min	Pctl(25)	Pctl(75)	Max
Control	0.110	0.591	−1.843	−0.087	−0.016	6.024
Treatment	0.113	0.589	−1.789	−0.088	−0.013	5.256
Incumbent	0.096	0.577	− 3.253	−0.108	−0.008	6.524

Demand Estimation

- Consumer Demand: [details](#) [estimates](#)
 - ▷ account for rich consumer preference heterogeneity
 - ▷ instrument for prices and product variety
IV: platform's rule enforcement + treatment assignment
 - ▷ estimate seller-specific unobserved “ability” ξ_j [details](#)
- Platform matching [details](#)
 - ▷ account for path-dependence and growth compounding in platform traffic allocation